

MARIA GRACIEL CRUZ CACERES

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SUMMARY

AI engineer building production multi-agent LLM systems at EY's Generative AI Hub, and an independent researcher with a first-author poster at the LXCW Workshop, CVPR 2026. At EY, I lead end-to-end delivery for major financial institutions — LangGraph pipelines, RAG over enterprise data, self-correcting code generation — owning architecture through production observability. My research investigates why multimodal models that handle physical causal reasoning degrade when the causal subject is a person's emotional state: a failure mode I identified during a Google Explore CSR fellowship and am now pursuing as a sustained research direction in causal affective VideoQA.

SKILLS

Generative AI & LLMs: LangChain, LangGraph, Azure OpenAI, RAG, FAISS, AI Agents

ML, CV & Multimodal: PyTorch, HuggingFace Transformers/Diffusers, BERT, Latent Diffusion

Languages & Data: Python, TypeScript, SQL, Scala, Pandas, NumPy, Spark, Databricks, PostgreSQL

Cloud & Tools: Azure, AWS (Certified Cloud Practitioner), Docker, OpenTelemetry, React, FastAPI, Pydantic, Pytest

Spoken Languages: Spanish (Native), English (Fluent — C1), Portuguese (Fluent)

EXPERIENCE

Ernst & Young — Generative AI Hub, Rio de Janeiro, Brazil

02/25 – Current

AI Engineer – Technology (*promoted Oct 2025 — ahead of EY's standard 12-month cycle*)

10/25 - Current

- **Credit Portability Engine:** As the technical lead embedded with a major financial institution's operations division — owning requirements through production — engineered a natural language-to-insight system using LangGraph and Azure OpenAI. The agent processes complex financial questions, autonomously generating and executing PostgreSQL queries to extract targeted financial data and producing self-correcting Python code for complex financial calculations. Implemented a sliding-window memory strategy — retaining the last 5 turns verbatim plus a running conversation summary refreshed every 5 turns — to preserve coherent long-horizon context within LLM token constraints. Architected the full stack including a React/TypeScript conversational interface deployed on Azure with OpenTelemetry observability.
- **End-to-End Automated Data Engineering:** Architected an autonomous multi-agent system (Azure OpenAI, LangGraph) that ingests raw client data, infers column types and semantics from file headers and sample rows when the data dictionary is absent or incomplete, and translates natural language business rules into executable Python code. Designed a sequential multi-agent self-correcting loop that iteratively generates, executes, and verifies code against client expectations — automating what was previously manual code generation from business rule specifications (a process requiring a dedicated engineering team over 2–3 months) down to a single non-specialist validating outputs through a structured human-in-the-loop interface. Containerized with Docker; delivered via a FastAPI/Pydantic backend and React/TypeScript frontend; Pytest suite covers core pipeline stages to prevent regressions across iterative development cycles.

AI Engineer – Technology

02/25 - 10/25

- **CodeJumper (JS/WSH to Python migration):** Designed a multi-stage LangGraph pipeline to automate legacy bank script migration. Parsed source scripts into an AST and used recursive traversal to decompose code into function- and class-level units; dispatched each unit in parallel to the LLM for translation. Implemented an LLM-as-judge quality gate scoring each output 1–10 on correctness and readability — units below threshold re-enter the pipeline with targeted error context, up to 3 iterations. Validated translated outputs via automated test execution before final assembly. Reduced conversion time by 75% (sprint-tracked against manual baseline); process now requires a single operator for output review.
- **RFP Intelligence (Strategic Bid Analyzer):** Engineered a RAG system (FAISS vector index over chunked documents) over thousand-page procurement contracts to automatically extract critical requirements and policy conflicts, distilling them into a structured one-page executive summary with an initial bid/no-bid signal — enabling rapid go/no-go decisions without full manual legal review. Deployed via a serverless Azure Functions backend integrated into Power Apps.

Ntt Data, Arequipa, Peru

10/24 – 02/25

Data Engineer – Technology

- Engineered an end-to-end data pipeline in Scala on Databricks to generate a critical bank-regulated report, translating an 80-field business requirements document — including complex dependencies, calculation logic, and regulatory constraints — into executable SQL and scalable Scala workflows across 22 unified tables.
- Implemented an AES-256 encryption module to securely automate the daily transfer of 20 highly sensitive reports.

Universidad Católica San Pablo, Arequipa, Peru

08/23 – 12/23

Teaching Assistant - Advanced Data Structures – Education

- Led class sessions and provided individual student support for one of Peru's most competitive CS programs (ABET-CAC accredited).

Research Assistant – Education

- Conducted research applying Transformer-based architectures to MHC-peptide binding classification for neoantigen detection in personalized cancer vaccine development.
- Extracted and compared architecture types, training strategies, and evaluation protocols across 50+ Transformer-based papers on MHC-peptide binding; systematic analysis contributed directly to both peer-reviewed publications.
- **Co-authored two peer-reviewed publications** (see Publications).

PUBLICATIONS & PRESENTATIONS

- Cruz Caceres, M.** "Asking Why: Causal Question Answering Over Human Affective and Intentional States in Multimodal Video." *LatinX in Computer Vision (LXCV) Workshop, CVPR 2026. (Poster Presentation)* 2026
Available at: <https://github.com/magrcruz/Asking-Why-Causal-Affective-QA>
- Cruz Caceres, M.** "Efficient Image Editing Using Compressed Latent Representations in Diffusion Models." *Khipu Latin American AI Conference, 2025. (Poster Presentation)* 2025
- Machaca, V., Goyzueta, V., **Cruz Caceres, M.**, et al. "Transformers Meets Neoantigen Detection: A Systematic Literature Review." *Journal of Integrative Bioinformatics, 2024.* 2024
- Machaca, V., Goyzueta, V., **Cruz Caceres, M.**, Túpac, Y. "Deep Learning and Transformers in MHC-Peptide Binding and Presentation Towards Personalized Vaccines in Cancer Immunology: A Brief Review." *PACBB 2023.* 2023

PROJECTS

Asking Why: Causal QA Over Human Affective States in Multimodal Video (Independent Research)

Identified a gap at the intersection of causal VideoQA and affective computing: existing models (NExT-QA, IntentQA) reason causally about physical events but degrade on latent emotional or intentional states. Proposed a two-stage methodology — benchmark gap analysis via SeViLA/Video-LLaVA evaluation on MELD and MIntRec2.0, followed by a retrieval-augmented causal reasoning prototype with dual output (timestamped evidence clips + natural language explanation). This work emerged directly from multimodal intent recognition research conducted under the Google Explore CSR fellowship— where evaluation of MAG-BERT and MISA architectures on MIntRec2.0 surfaced a systematic failure mode in causal reasoning over affective states that existing benchmarks did not capture. Accepted for poster presentation at LXCV Workshop, CVPR 2026.

Link: <https://github.com/magrcruz/Asking-Why-Causal-Affective-QA>

Image Editing Using Diffusion Models (Undergraduate Thesis)

Adapted a compressed latent diffusion model (Würstchen) to enable fast text-guided image editing by injecting spatial features from the original image into the denoising process. Achieved ~10x faster inference than standard baselines (~24s vs. 2–4 minutes) while maintaining comparable image quality (evaluated via CLIP Score and CLIP Similarity against source prompts and reference images). Benchmarked against Imagic, Plug-and-Play Features, and HiPer.

EDUCATION

Universidad Católica San Pablo, Arequipa, Peru

Bachelor of Computer Science, Concentration: Artificial Intelligence. GPA 4.0

- **Thesis:** Efficient Image Editing using Compressed Latent Representations in Diffusion Models
- **Relevant Coursework:** Artificial intelligence, Cloud computing, Big Data, Advanced Data Structures, Software Engineering.

Pontifícia Universidade do Rio de Janeiro, Rio de Janeiro, Brazil

Bachelor of Computer Science (Study abroad)

- **Relevant Coursework:** Data Science, Machine Learning and Human-Computer Interaction.

AWARDS AND RECOGNITIONS

- **CVPR Full Registration Grant** - LatinX in Computer Vision (LXCV) Workshop, Poster Presenter 2026
- **Khipu AI Conference Scholarship** -Latin American AI Conference (image diffusion research) 2025
- **Google Explore CSR Research Fellowship** — Latin American Undergraduate AI Research Program, Google 2024
(selected participant; conducted multimodal intent recognition research on MIntRec2.0 using MAG-BERT and MISA architectures; identified failure mode in causal affective reasoning that became the foundation of CVPR 2026 LXCV work)
- **Finalist** - II Women's Programming Competition MFP, Brazilian Computer Society 2024
- **CADE-IPAE Invited Participant** - National Meeting for Young Leaders 2024
- **1st Place, CS Program** - Universidad Católica San Pablo 2019–2024

VOLUNTEER

Association for Computing Machinery (ACM), Arequipa, Peru

01/23 – 12/23

ACM Student Chapter Member – Technology

- Led logistics for Techsuyo, hosting Peru's first Google Developer Summit with 200+ attendees.
- Prepared materials for CEP 2023 School Programming Contest, quadrupling the number of participating schools.